

Sukhrob Ilyosbekov

Machine Learning Engineer · Computer Vision & LLM Systems

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SUMMARY

Machine learning engineer with three arXiv papers in computer vision and multimodal learning and nine years of production software. Ships deep-learning and LLM systems end to end: PyTorch models behind FastAPI, clinical imaging inside OHIF, a Claude tool-use agent dispatching freight. Applied-math background.

TECHNICAL SKILLS

Deep Learning & CV: PyTorch, TensorFlow, CNNs, Vision Transformers, DINOv3, EfficientNet, YOLO, contrastive learning, GradCAM++, OpenCV

LLM Systems: Claude/GPT/DeepSeek APIs, tool-use agents, MCP, RAG, vision-language models, Hugging Face

Math & Modeling: Probability and statistics, linear algebra, optimization, time-series forecasting, feature engineering, A/B testing, error analysis

Languages: Python, C#, C++, TypeScript, Kotlin

Data & Serving: NumPy, pandas, scikit-learn, FastAPI, PostgreSQL, SQL Server, MongoDB, Redis, OHIF/DICOM

Infrastructure: CUDA, mixed-precision and multi-GPU training, model serving and drift monitoring, Docker, Kubernetes, GitHub Actions, AWS, Azure

Software: ASP.NET Core, SignalR, Blazor, Node.js, Bun, NestJS, React, Next.js, Angular

EDUCATION

Northeastern University, M.S. in Computer Science, GPA 4.0, Boston, MA

Jan 2024 – May 2026

Suffolk University, B.S. in Computer Science, Cum Laude, Boston, MA

Sep 2021 – May 2023

Honors: National Physics Olympiad, 1st Place (Uzbekistan) · Associate Member, Sigma Xi · University Coding Competition Winner, TUIT

PROFESSIONAL EXPERIENCE

Senior Software Engineer | [EmTech Care Labs](#) | Portland, ME

Jan 2025 – Present

- Decline-risk model over daily activity and care-log data for Alzheimer's residents; 0.81 AUC on held-out data.
- NLP pipeline extracting symptoms, medications, and care events from free-text clinical notes; 0.90 F1.
- ML serving on de-identified data under HIPAA: nightly retraining, drift monitoring, encrypted PHI, audit logging.
- Lead engineer on the platform itself (TypeScript monorepo, AWS); passed an outside HIPAA readiness review with no critical findings.

Senior Software Engineer | [AllFactors](#) | Santa Clara, CA

Oct 2021 – Dec 2024

- Property-valuation and market-forecast models over 2M listings in 40+ metros; median error 9% to 5% against the rules-based baseline.
- Real-time feature pipeline on SignalR and SQL Server change tracking; nightly retraining on the day's transactions.
- Served predictions into Blazor WebAssembly dashboards, with automated tests on the data and prediction flows.

Earlier: Software Engineer, Smart Meal Service, Russia (2020 – 2021), C#/.NET control software for food kiosks and a robotic cashier. Game Developer, Pentalight Technology, Malaysia (2020 – 2021), Unity networking and hand tracking for a VR simulation. Freelance Developer (2019 – 2020), Python NLP and .NET. Game Developer, EC Dev Team, Uzbekistan (2016 – 2019).

SELECTED AI/ML PROJECTS

MorphoCLIP

[GitHub](#) | [Paper](#)

PyTorch, DINOv3, BioClinical ModernBERT, contrastive learning, CPJUMP1

- Contrastive image-text model matching Cell Painting images of drug- and gene-perturbed cells to plain-language treatment descriptions.
- Frozen DINOv3 and BioClinical ModernBERT backbones with trained projection heads; trains on one consumer GPU.
- Evaluated on CPJUMP1 (51 plates, 3M+ images). First author; led design, training, and evaluation. On arXiv.

Localize, Don't Beautify

<https://arxiv.org/abs/2608.02841>

Image-editing APIs, inpainting, ArcFace, prompt engineering

- Benchmarked prompt steering, masked compositing, and inpainting across six commercial image editors on 196 cosmetic-surgery preview edits.
- Scored identity preservation (ArcFace) and edit localization; masked compositing localized best. Sole author. On arXiv.

MelanomaNet

[GitHub](#) | [Paper](#)

PyTorch, EfficientNet V2, focal loss, GradCAM++, OpenCV, ISIC 2019

- EfficientNet V2 with focal loss over nine ISIC 2019 categories, 25K images; 0.86 weighted F1.
- GradCAM++ attention scored against ABCDE dermatology criteria measured from the lesion mask. Sole author. On arXiv.

Med Image Scanner

<https://github.com/suxrobgm/med-image-scanner>

FastAPI, PyTorch, OpenCV, OHIF Viewer, DICOM, Next.js

- Pulls X-ray, CT, and MRI from hospital imaging systems over DICOM and runs PyTorch detectors: pneumonia on chest X-ray, hemorrhage on head CT.
- Renders predictions as overlays inside the OHIF viewer; on-the-fly de-identification, audit logging, and role-based access.

LogisticsX

<https://logisticsx.app> | [GitHub](#)

Claude API, MCP, .NET 10, EF Core, SignalR, Angular 21, Kotlin Multiplatform, PostgreSQL

- Claude tool-use agent that matches freight loads to trucks, checks federal hours-of-service, and plans multi-stop routes without a dispatcher.
- Full product around it: multi-tenant .NET backend, Angular portals, Kotlin Multiplatform driver app, load-board and telematics integrations.

Bookshelf Scanner

<https://github.com/suxrobgm/bookshelf-scanner>

YOLO, Moondream2 VLM, llama.cpp, FastAPI, Angular

- YOLO segmentation isolates each book spine; Moondream2 reads title and author; Angular UI for corrections and export. Course Outstanding Project Award.

OGStack

<https://ogstack.dev> | [GitHub](#)

Next.js 16, ElysiaJS, Bun, Satori, Cloudflare R2, Stripe

- Multi-model layer (Claude, GPT, DeepSeek) that reads a page's branding and generates its social-share image; 10K+ images in beta, served from a global CDN.